

Standing Waves and Electrons Around the Nucleus

Objectives:

With the development of quantum mechanics and experimental findings (such as the two slits diffraction of electrons), it was found that the orbiting electrons around a nucleus could not be fully described as particles, but need to be explained by the wave-particle duality. In this sense, the electrons do not orbit the nucleus in the sense of a planet orbiting the sun, but instead exist as **standing waves**. The lowest possible energy an electron can take is therefore analogous to the fundamental frequency of a wave on a string. Higher energy states are then similar to harmonics of the fundamental frequency.

We will study the vibrations of a tense string. You will see that this is a good way to get at waves. In particular, we are going to investigate standing waves, which are an indication of resonance. Standing waves quite literally oscillate up and down but do not go anywhere (unlike traveling waves or pulses).

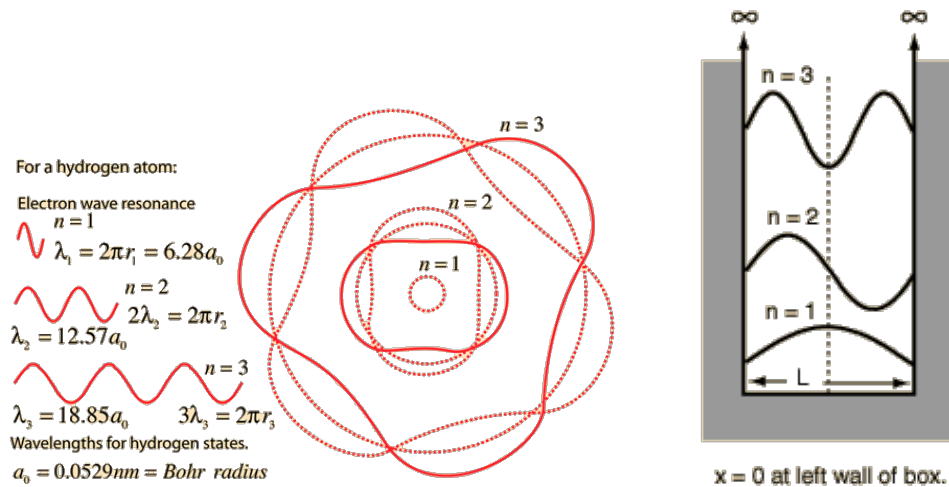


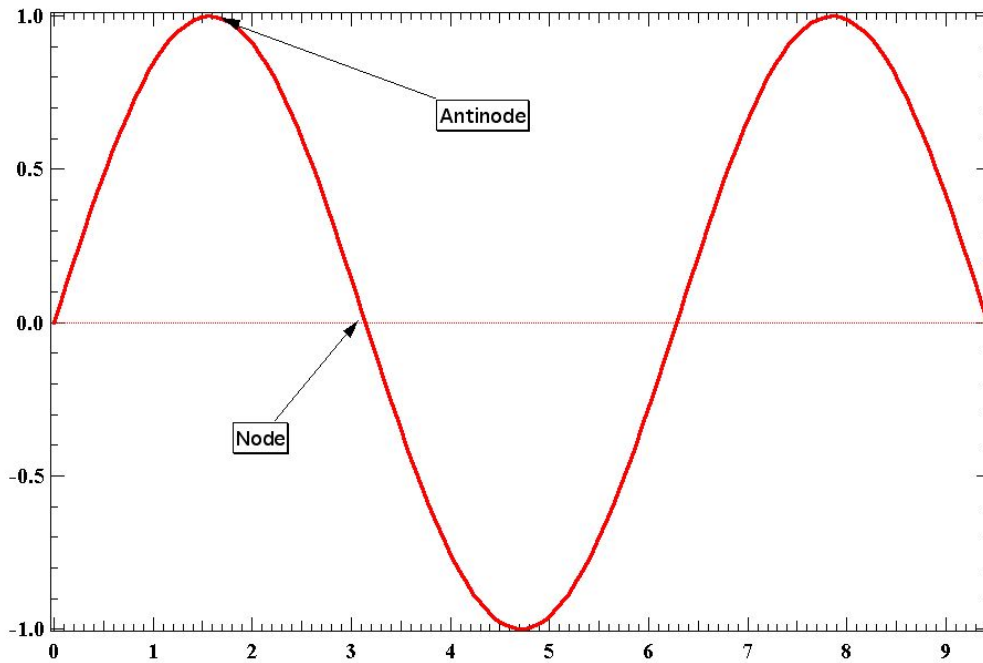
Figure: Electron standing waves on Hydrogen atom orbits (LEFT). Particle in a box problem solutions for the Schrodinger equation (RIGHT).

Theory:

Any wave will have a *frequency* (f) and a *wavelength* (λ). These two quantities are related to the traveling speed of the wave by the equation

$$v = f\lambda \quad (1)$$

Now, when you generate such a traveling wave on a string, say, and it hits the end (the support), it will reflect. It will then interfere with the next wave coming down the line and, if the conditions are right, we will create a standing wave. What does the standing wave look like? Well, look at the next graph.



You will notice that it look like a sine wave. That is because it is! You will also notice that there are no vibrations at the extremes (0 and $3\pi=9.4247$, in this case). That is because that's where the string is tied to the supports. When parts of the standing wave do not move up and down we call those points *nodes*. The points of greatest oscillation are the *antinodes*.

Now, what are the conditions necessary to make a standing wave? Well, we need to match the length of the wave (λ) and the length of the string (L). The condition is

$$L = \frac{n\lambda}{2} \quad (2)$$

The letter n in the equation denotes the normal mode of vibration. It is a positive integer. Each n will show you a differently-shaped standing wave, with greater n having more nodes. For instance, in the graph from the previous page, $n = 3$, and there are 4 nodes. The $n=4$ wave would have 5 nodes, and so on.

The last theoretical point has to do with the relationship between the tension in the string and the velocity. We know that the speed v of the traveling wave is given by

$$v = \sqrt{\frac{T}{\mu}} \quad (3)$$

where T is the tension in the string and μ is the mass per unit length of the string.

Combining Equations (1) and (3), we get

$$f\lambda = \sqrt{\frac{T}{\mu}}, \text{ or } (f^2\mu)\lambda^2 = T \quad (4)$$

We write it this way because, for Experiment 4, the frequency f will be fixed and μ is also a constant. So T is a linear function of λ^2 .

3. Procedure

First, make sure that the mechanical wave driver is secure on a clamp and connected to the function generator. The string should be perfectly horizontal, and slung over a pulley. At the end of the string there should be a weight hanger. If the string is frayed or too short, replace the string. Before you start, weigh the whole string (but not the hanger), and measure its entire length $\mathcal{L}$. Get the mass per unit length $\mu = m/\mathcal{L}$.

Experiment 1

Add weights so that the hanging mass (including the hanger) is 100 grams. Then measure the string length from the wave driver to the pulley contact (**not** down to the hanger). The function generator should be connected to the mechanical wave driver. Turn on the function generator, making sure that it is set to generate sine waves, and turn the output amplitude to the maximum setting.

Adjust the frequency of the function generator until you get the standing wave mode with $n=2$. Use Equation (2) to get the wavelength. Now use equation (3) to calculate a predicted wave speed v . Compare this with the result obtained by using your measured values of f and λ in Equation (1).

Experiment 2

Repeat the procedure for Experiment 1, adjusting the frequency until you see the modes with $n=2, 3, 4, 5, 6$. Record the data in the table in the Data section. Compare the predicted and measured values for the wave speed in the string. Plot f^2 (in the y-axis) vs. $1/\lambda^2$ (in the x axis). Note that the slope of your line should be T/μ : compare what you actually get from the graph to the value of tension/ μ .

Experiment 3

Repeat the procedure of Experiment 2 again using a hanging mass of 200 g. Once again record the data in the table in the Data section and compare the two values for the wave speed in the string. Plot f^2 vs. $1/\lambda^2$. Note that the slope of your line should again be T/μ : compare what you actually get from the graph to the value of tension/ μ . How do your results differ from Experiment 2?

Experiment 4

Set up the apparatus as in Experiment 1, with just 50 g on the end of the string, and adjust the frequency so that the $n=6$ standing wave is present. Now keep the frequency the same and add weights until you see the modes $n=3, 4, 5, 6$. Do not use a total mass of more than 250 g. Record the data in the table in the Data section. Use your data to calculate T from Equation 4 for each case. Compare these with the measured values $T=mg$.

Now use the table data to plot the measured T in the y-axis vs. λ^2 in the x-axis. What should be the slope? Do a fit (or linear regression) using the computer. Compare the slope of your graph to the slope predicted by the equations.

Experiment 5

According to the theory the distance between nodes in the standing wave is equal to $\lambda/2$. Since the end of the string attached to the driver is moving and the pulley end is not fixed, neither is a true node although in the previous experiments we treated both ends as nodes. Repeat a few of the trials from Experiments 2 and 3, this time measuring the distance between two nodes on the string (not at either end). Calculate the wave speed using the values of λ obtained this way. Does this method seem better?

4. Data

Experiment 1

Mass of the string: _____ kg; and its total length $\mathcal{L}$ was _____ m

So μ = _____ kg/m

Effective length L = _____ m, and so λ = _____ m

Frequency for the $n=2$ mode: f = _____ Hz

Measured tension in the string $T = mg$ = _____ N

Calculated wave speed in the string from Eq. (3) v = _____ m/s

Measured wave speed from Eq. (1) v = _____ m/s

Experiment 2: Varying f

Hanging mass: _____

n	Measured $T=mg$ (N)	Predicted $v = \sqrt{\frac{T}{\mu}}$	Measured f (Hz)	Wavelength λ (m)	Measured $v=f\lambda$
2					
3					
4					
5					
6					

Experiment 3: Varying f

Hanging mass: _____

n	Measured $T=mg$ (N)	Predicted $v = \sqrt{\frac{T}{\mu}}$	Measured f (Hz)	Wavelength λ (m)	Measured $v=f\lambda$
2					
3					
4					
5					

6					
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Experiment 4: Varying T

Frequency $f = \rule{1cm}{0.4pt}$ Hz

n	Hanging mass (kg)	Tension mg (N)	Wavelength λ (m)	Calculated T (N)
3				
4				
5				
6				

5. Questions

1. What are the most likely sources of error? Which of these is the largest?

2. What is the longest possible wavelength λ for a standing wave in terms of the string length L ?

3. By how much should we increase the tension T to increase the wave speed v by a factor of 3? How much should we change the mass per unit length μ to decrease the wave speed v by a factor of 2?

4. Why do you think the mass per unit length μ is important? Why is it in the equation for speed? Try and give *physical* reasons.